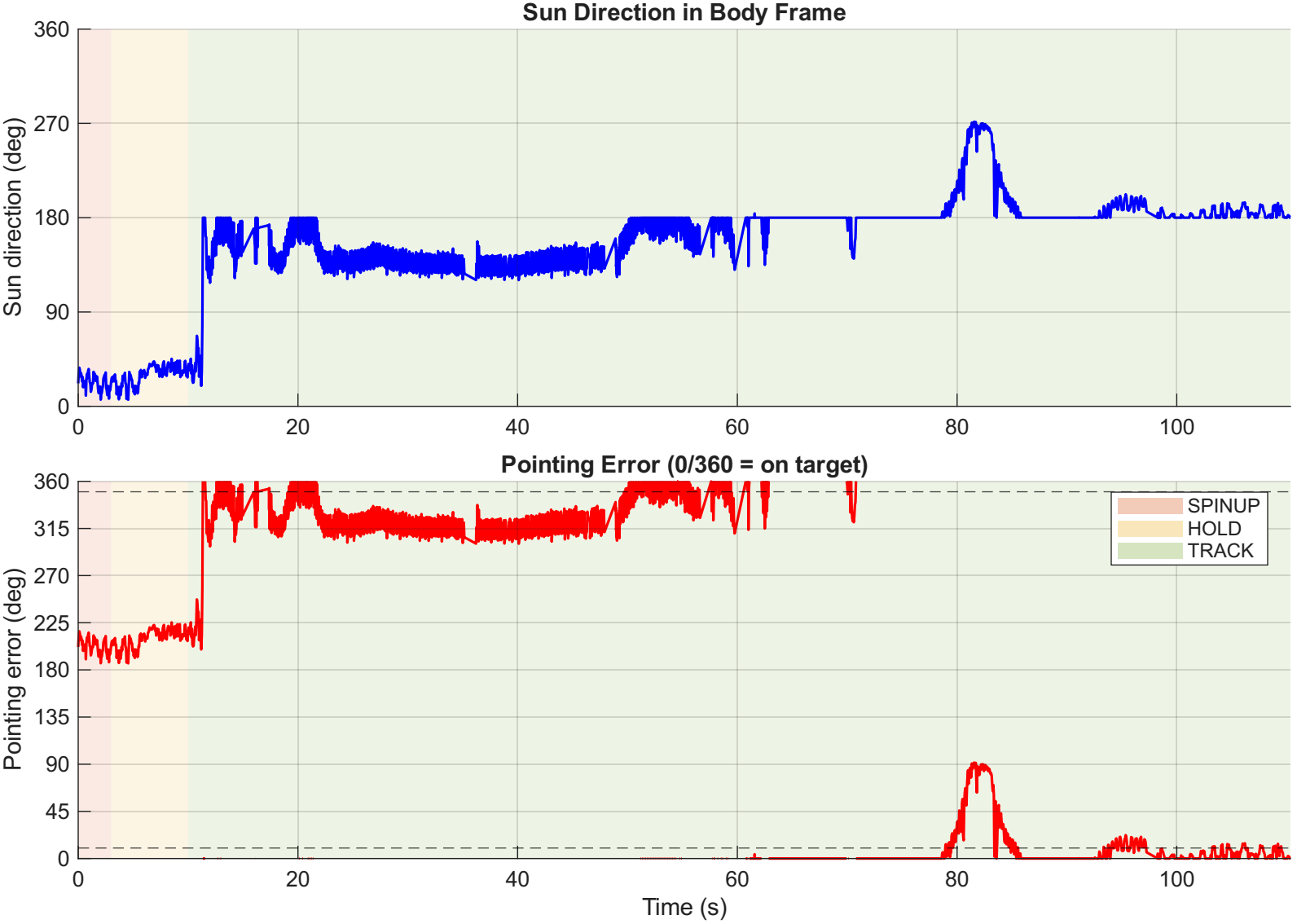
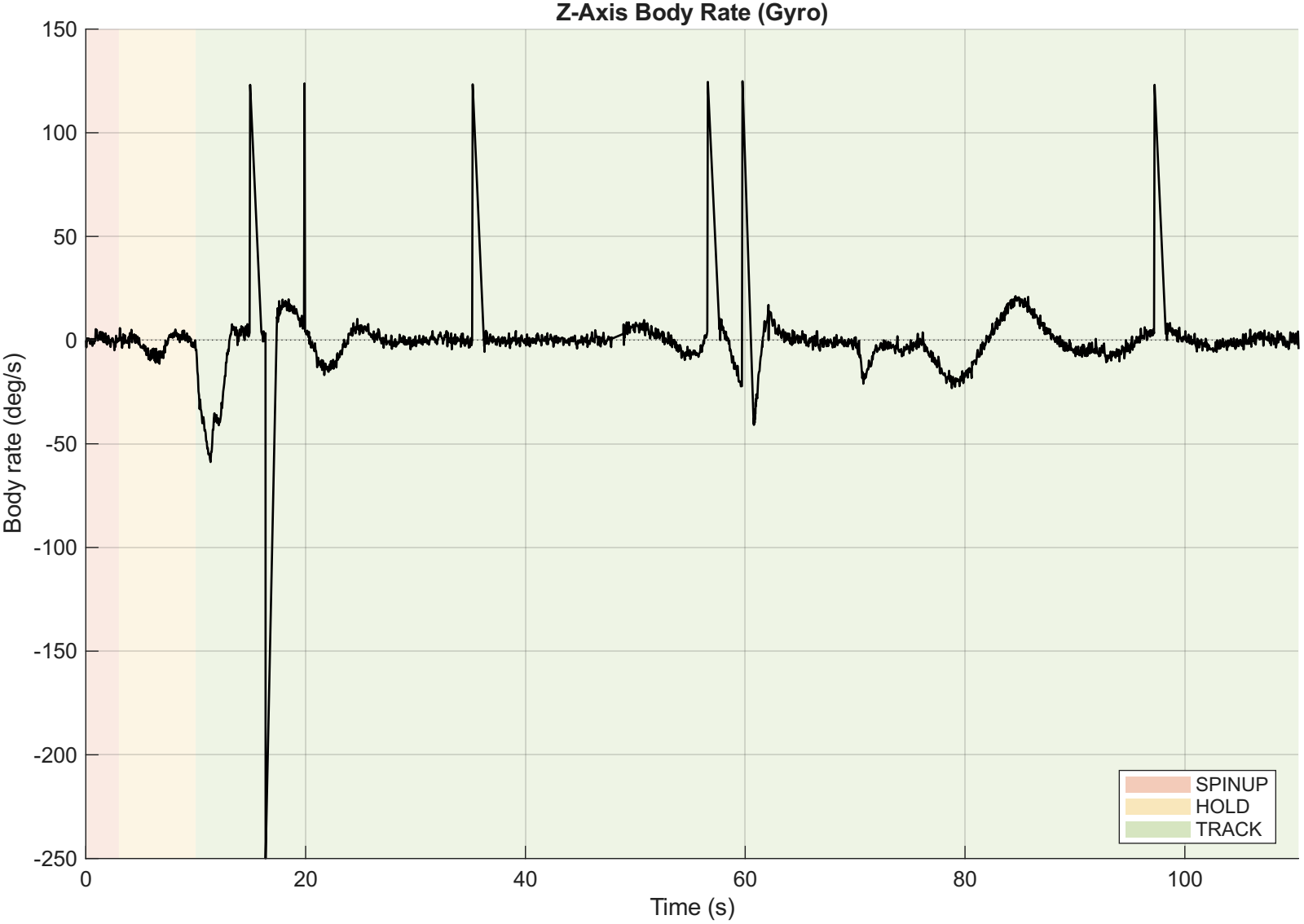
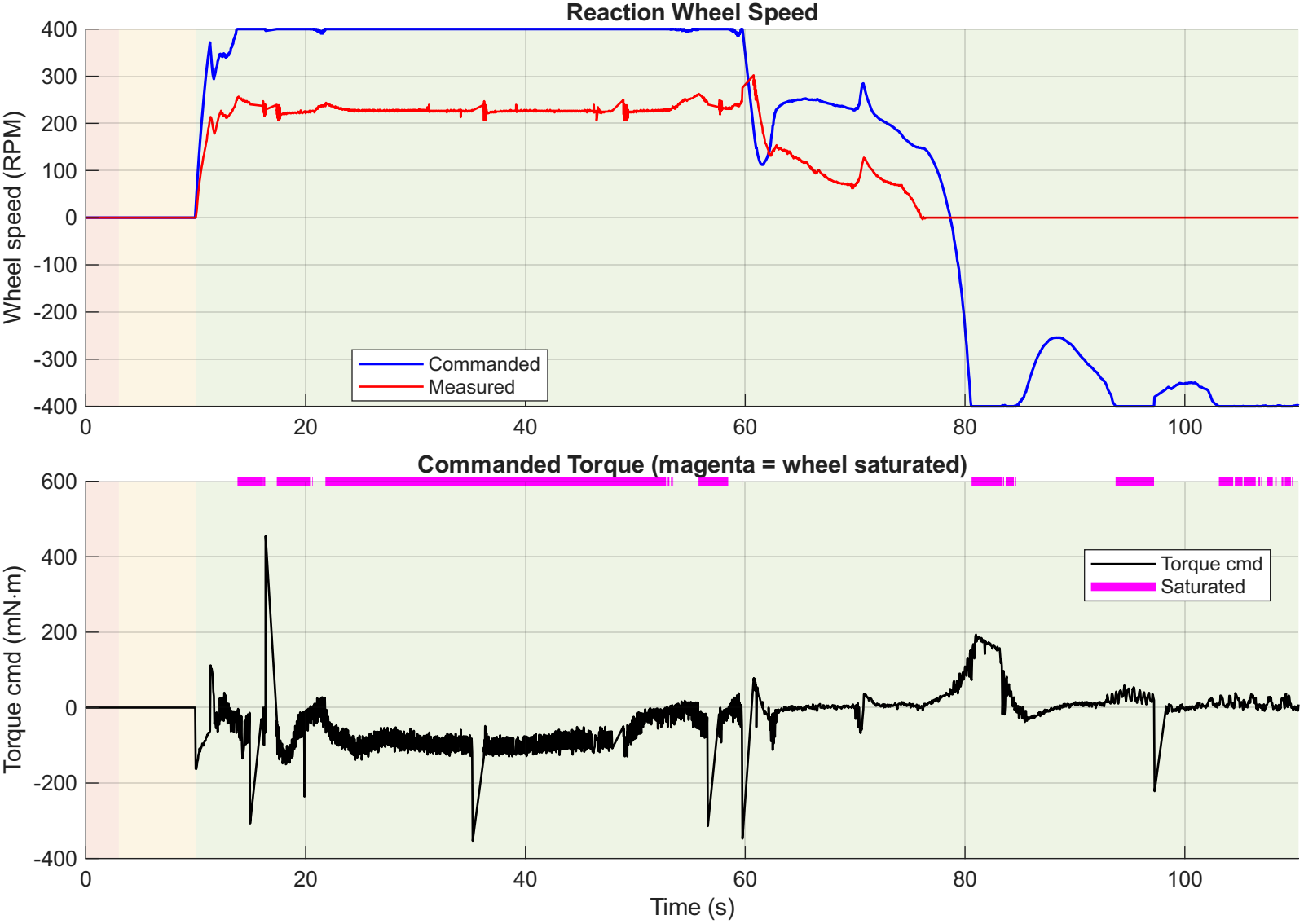
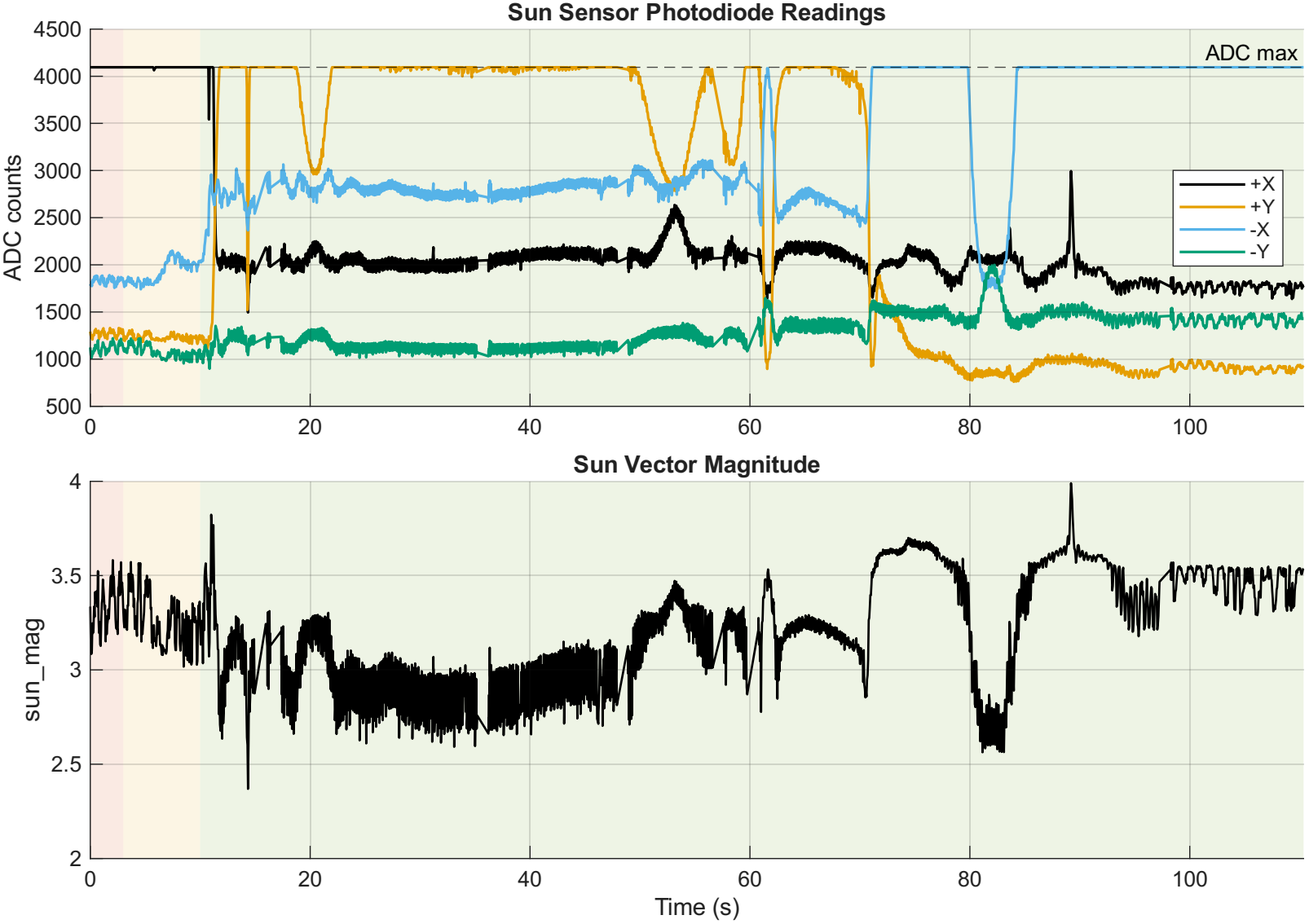
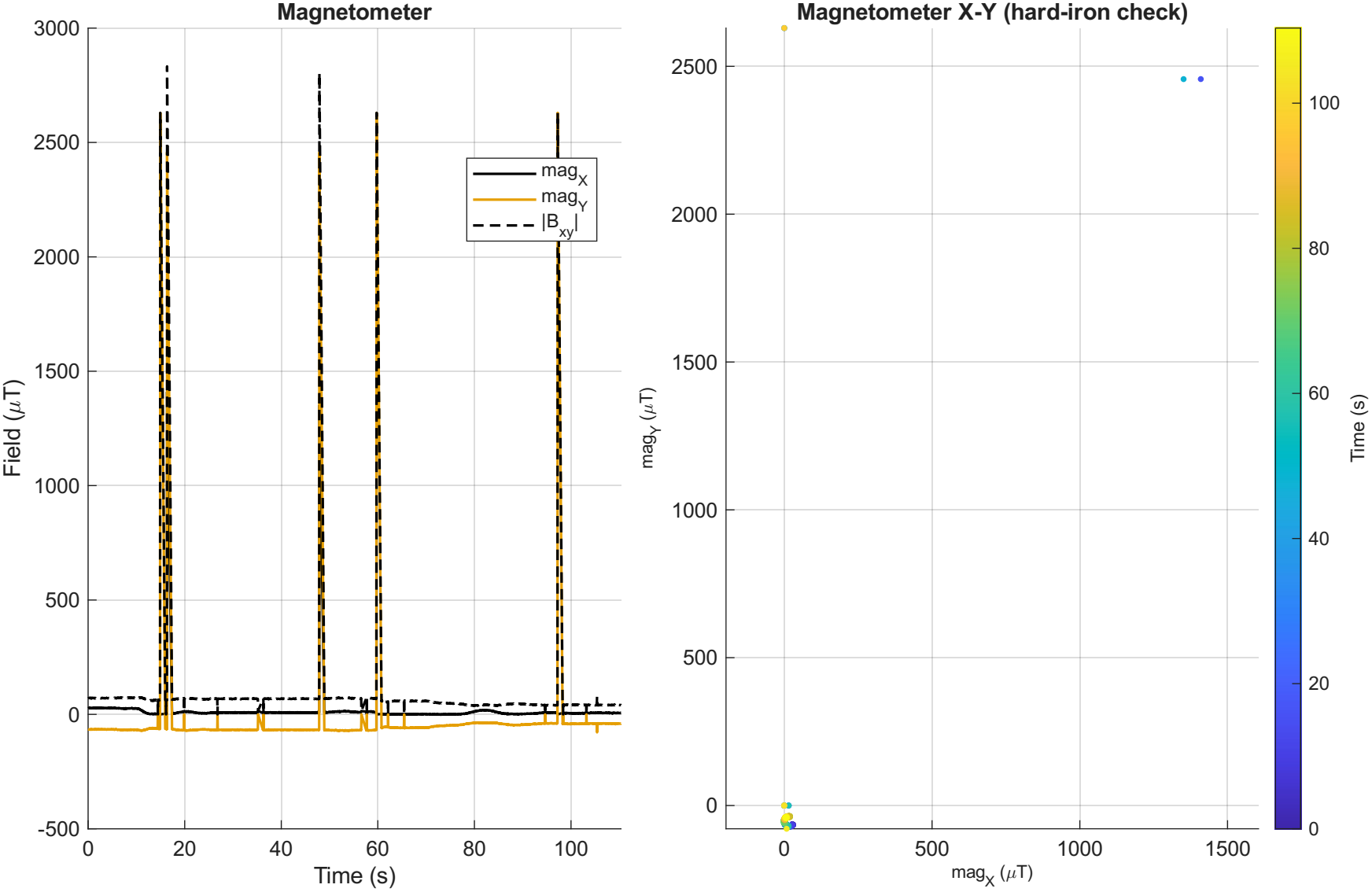
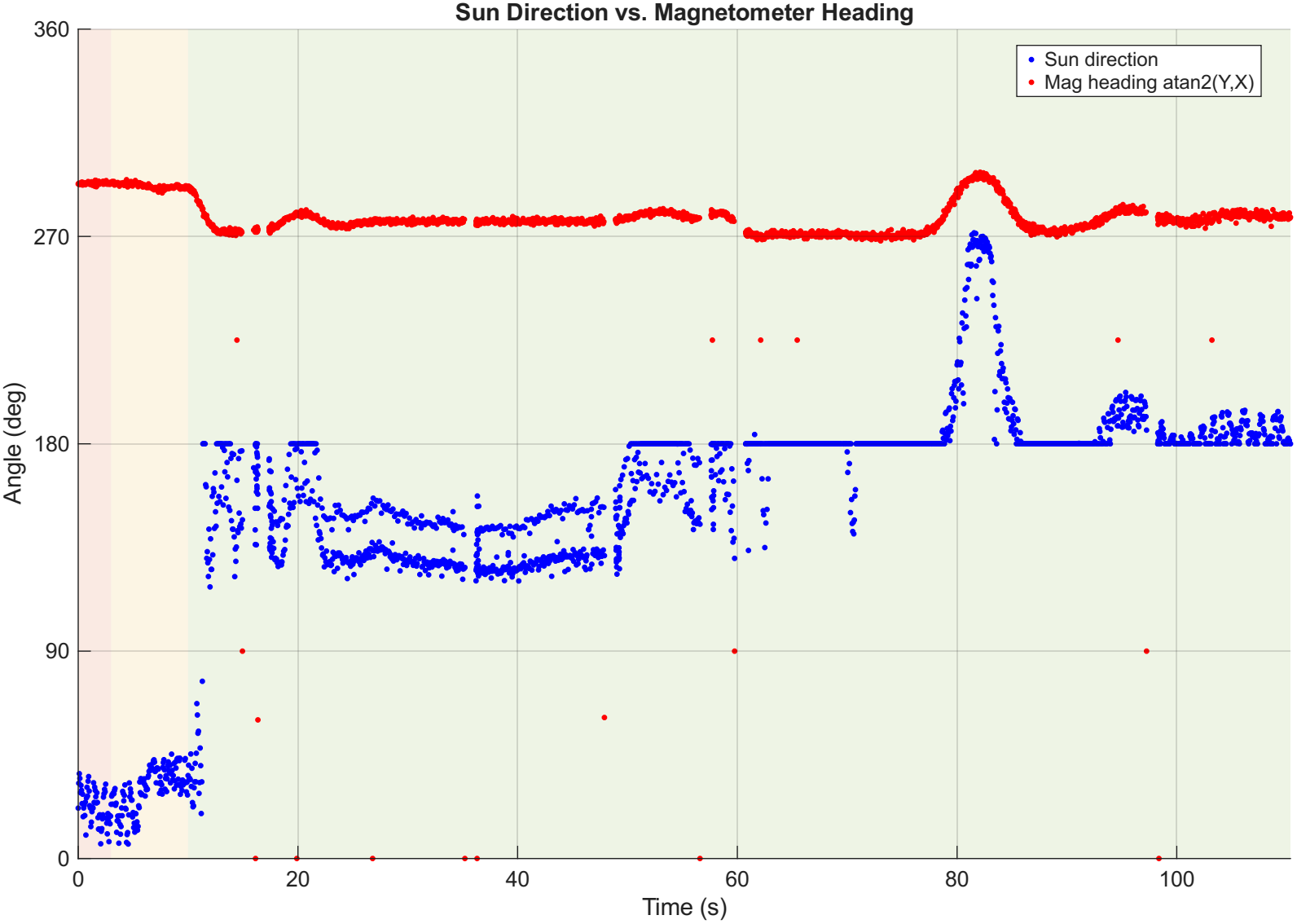

Sun Direction in Body Frame
Sun direction (deg)
360
270
180
90
0
0
20
40
60
80
100
Pointing Error (0/360 = on target)
Pointing error (deg)
360
315
270
225
180
135
90
45
0
SPINUP
HOLD
TRACK
Time (s)

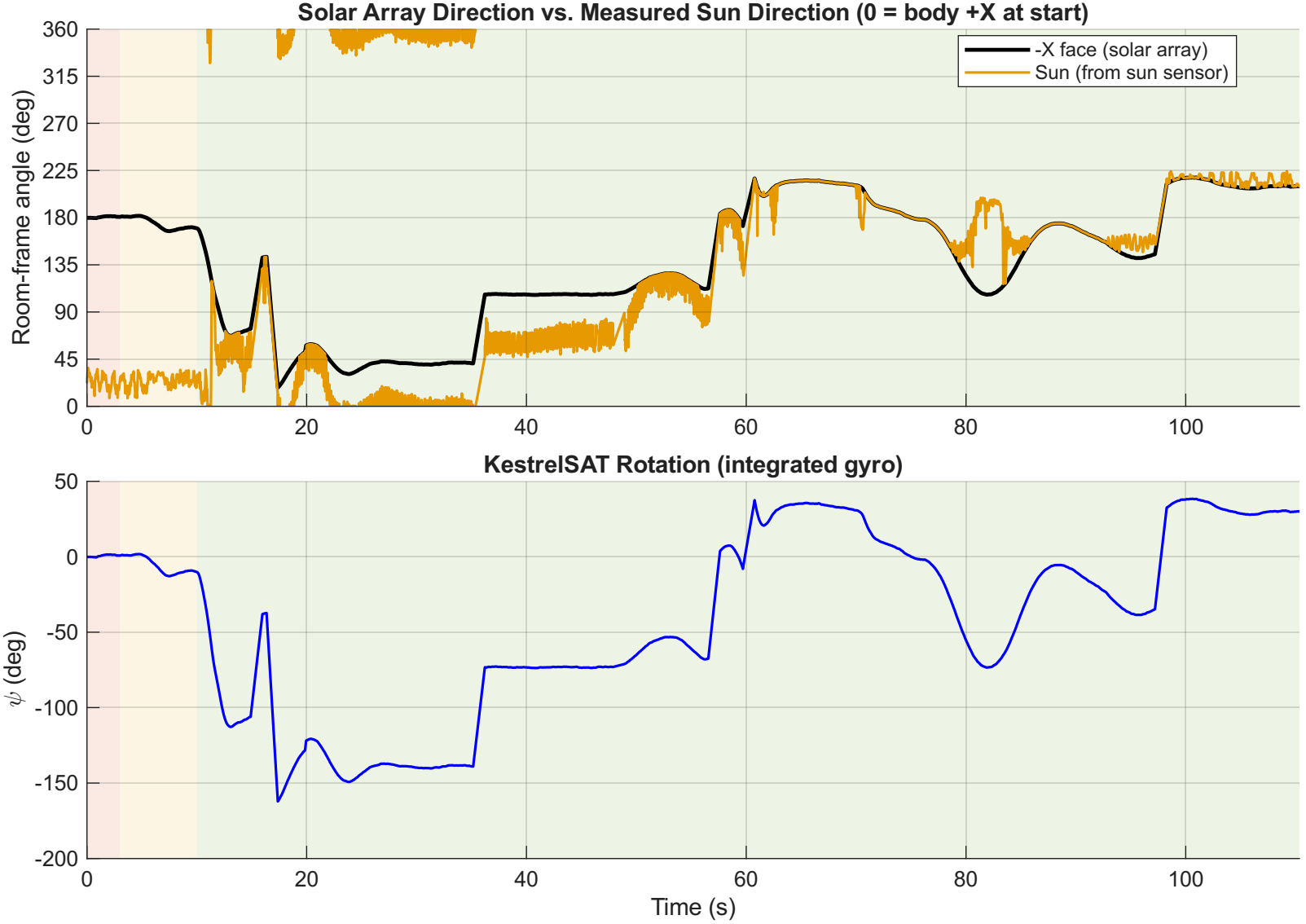

Z-Axis Body Rate (Gyro)
Body rate (deg/s)
Time (s)
150
100
50
0
-50
-100
-150
-200
-250
0
20
40
60
80
100
SPINUP
HOLD
TRACK

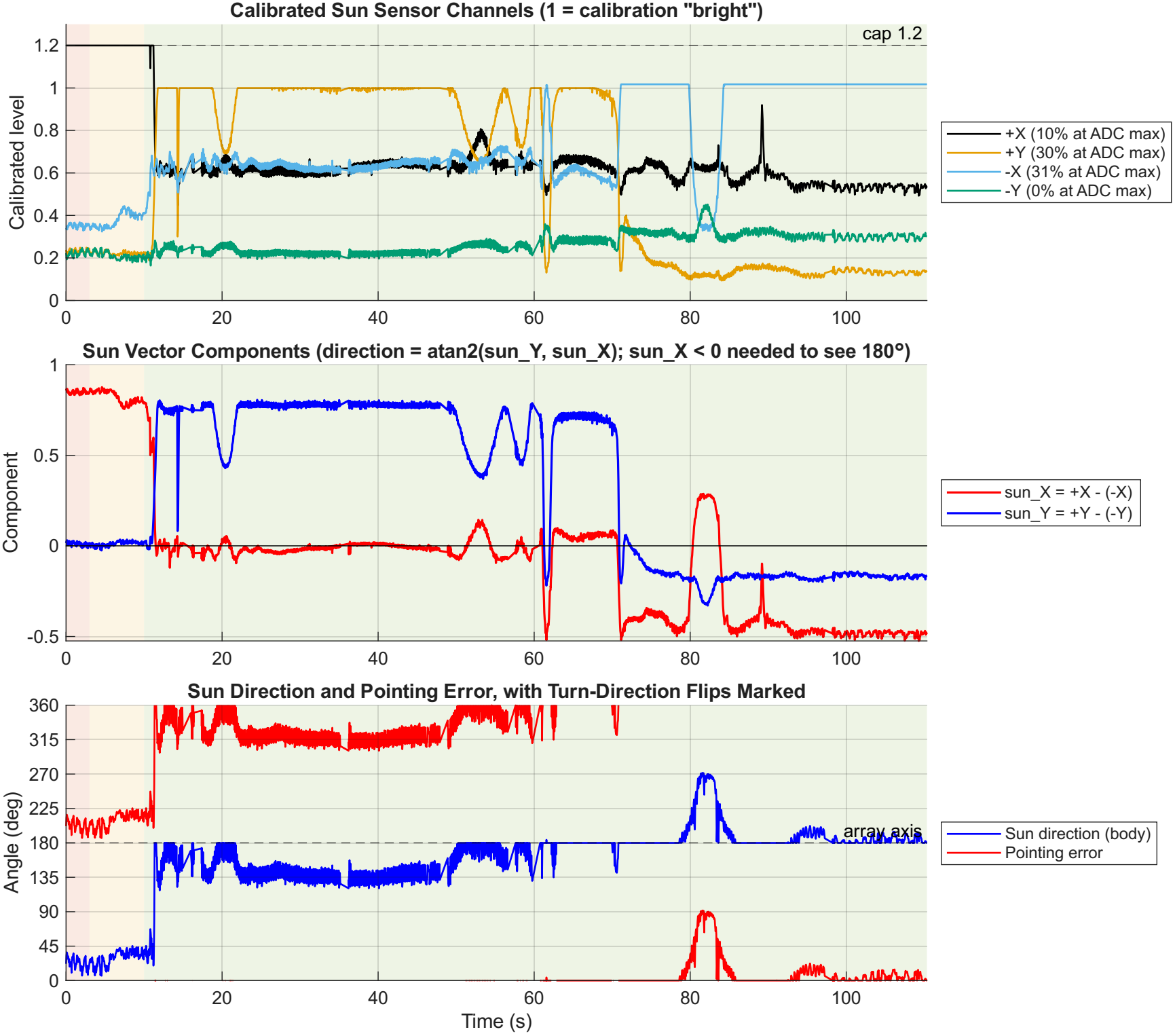

Reaction Wheel Speed
Wheel speed (RPM)
400
300
200
100
0
-100
-200
-300
-400
0
20
40
60
80
100
Commanded
Measured
Commanded Torque (magenta = wheel saturated)
Torque cmd (mN·m)
600
400
200
0
-200
-400
Torque cmd
Saturated
Time (s)

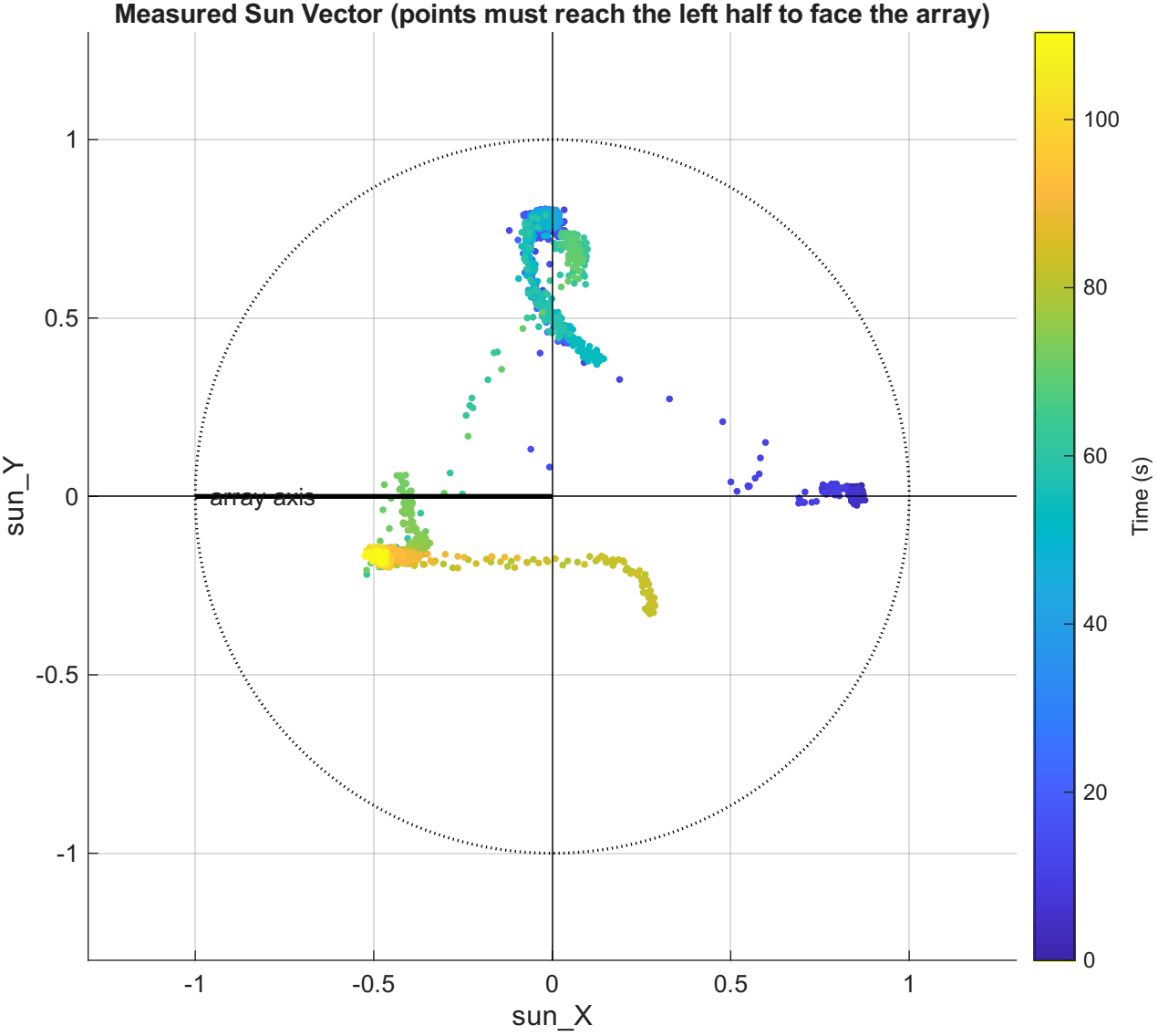

Sun Sensor Photodiode Readings
ADC max
ADC counts
4500
4000
3500
3000
2500
2000
1500
1000
500
+X
+Y
-X
-Y
0
20
40
60
80
100
Sun Vector Magnitude
sun_mag
4
3.5
3
2.5
2
Time (s)

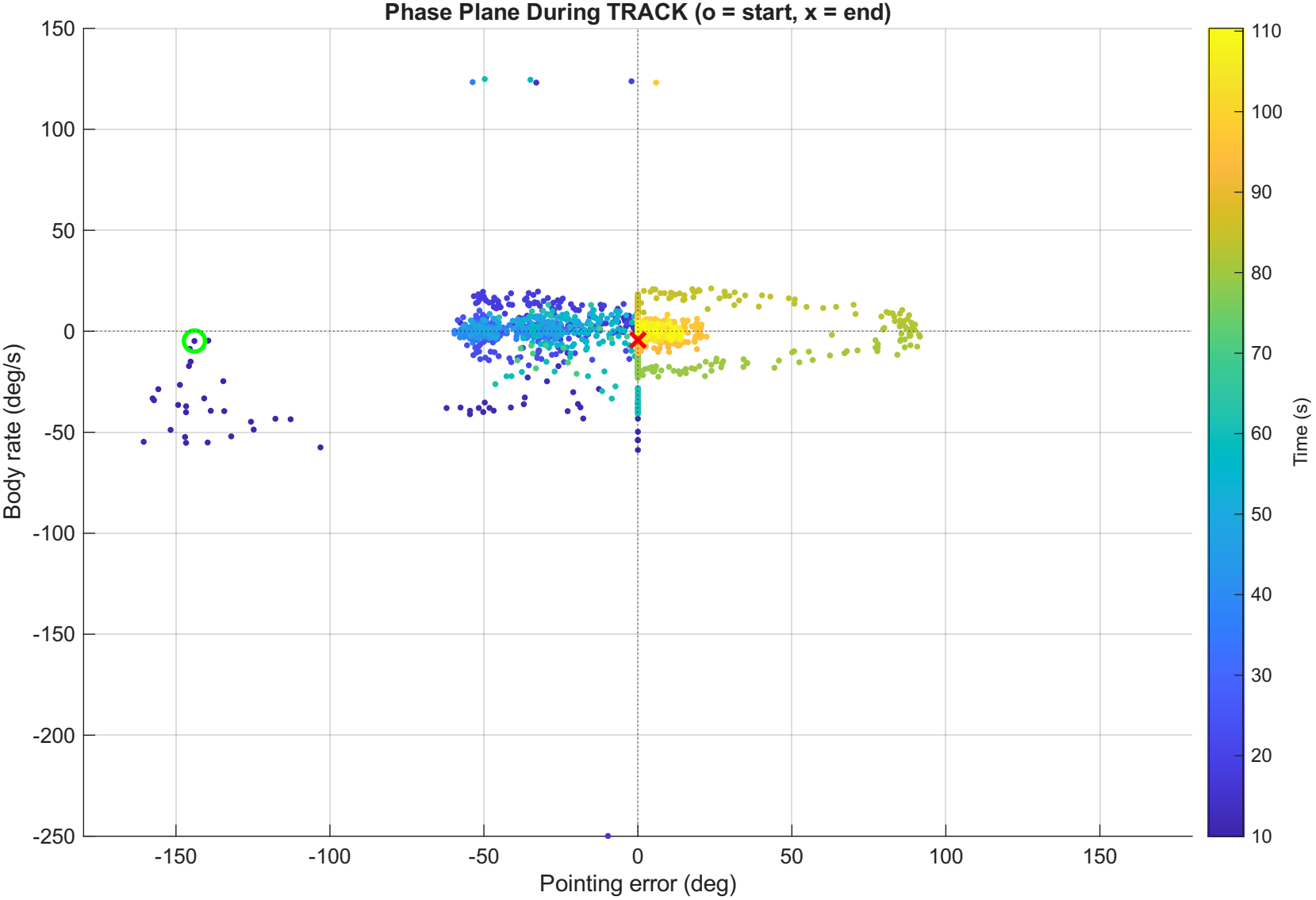

Magnetometer
Field (μT)
Time (s)
mag_X
mag_Y
|B_xy|
3000
2500
2000
1500
1000
500
0
-500
0
20
40
60
80
100
Magnetometer X-Y (hard-iron check)
mag_Y (μT)
mag_X (μT)
Time (s)
2500
2000
1500
1000
500
0
0
500
1000
1500
100
80
60
40
20
0

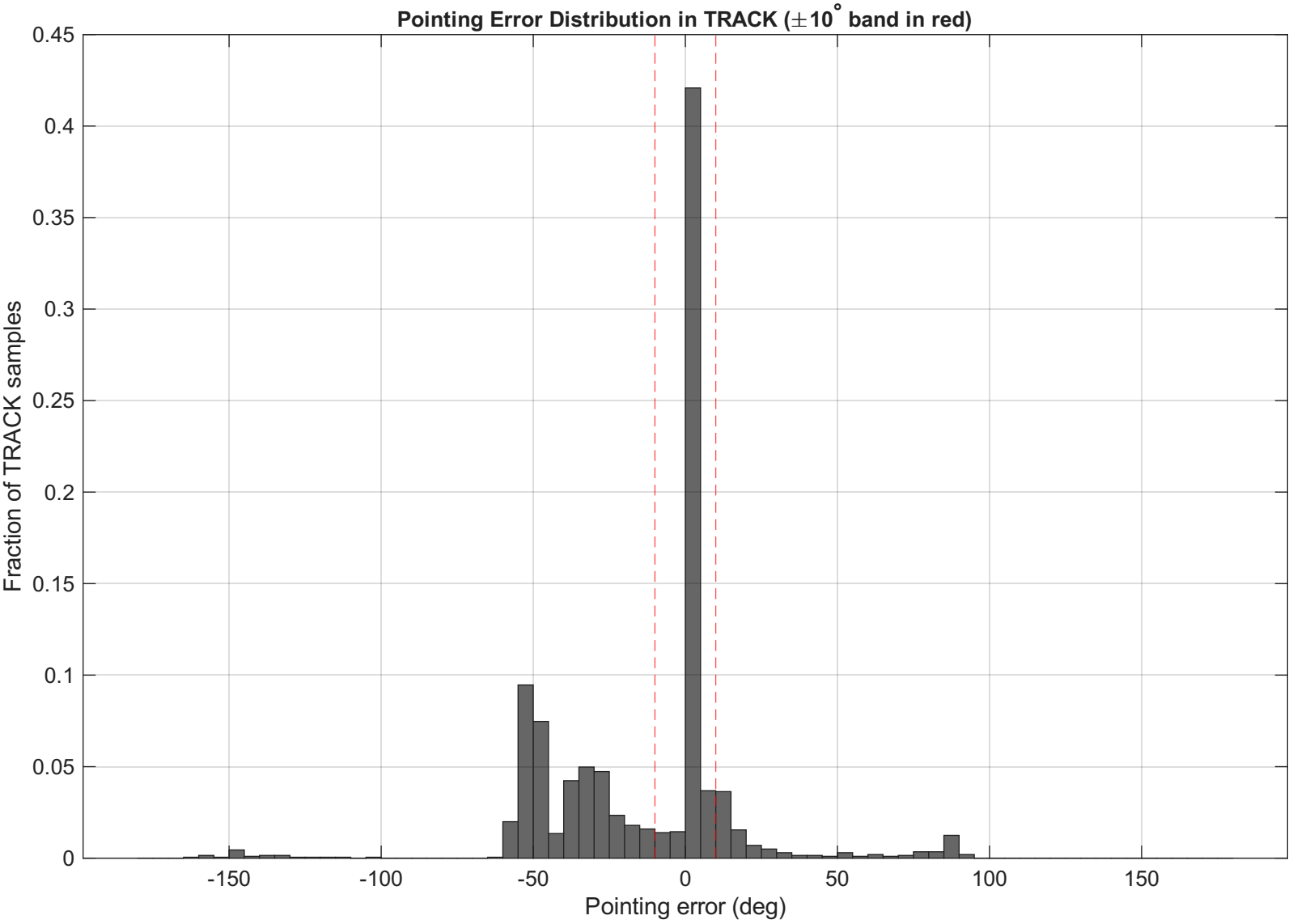

Sun Direction vs. Magnetometer Heading
Sun direction
Mag heading atan2(Y,X)
Angle (deg)
360
270
180
90
0
0
20
40
60
80
100
Time (s)


Solar Array Direction vs. Measured Sun Direction (0 = body +X at start)
-X face (solar array)
Sun (from sun sensor)
Room-frame angle (deg)
360
315
270
225
180
135
90
45
0
0
20
40
60
80
100
KestrelSAT Rotation (integrated gyro)
ψ (deg)
50
0
-50
-100
-150
-200
Time (s)


Calibrated Sun Sensor Channels (1 = calibration "bright")
cap 1.2
Calibrated level
+X (10% at ADC max)
+Y (30% at ADC max)
-X (31% at ADC max)
-Y (0% at ADC max)
Sun Vector Components (direction = atan2(sun_Y, sun_X); sun_X < 0 needed to see 180°)
Component
sun_X = +X - (-X)
sun_Y = +Y - (-Y)
Sun Direction and Pointing Error, with Turn-Direction Flips Marked
Angle (deg)
array axis
Sun direction (body)
Pointing error
Time (s)


Measured Sun Vector (points must reach the left half to face the array)
array axis
sun_Y
sun_X
Time (s)
1
0.5
0
-0.5
-1
-1
-0.5
0
0.5
1
100
80
60
40
20
0


Phase Plane During TRACK (o = start, x = end)
Body rate (deg/s)
Pointing error (deg)
Time (s)
150
100
50
0
-50
-100
-150
-200
-250
-150
-100
-50
0
50
100
150
110
100
90
80
70
60
50
40
30
20
10


Pointing Error Distribution in TRACK (±10° band in red)
Fraction of TRACK samples
Pointing error (deg)
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
-150
-100
-50
0
50
100
150